kader Academy

Machinery Safety Specialist (CE)

Tab 4: Risk reduction and standards
Module 1: Risk reduction principles

Tab4-1/5 20220802

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Tab1-1/5-Sh.2

Risk assessment in accordance with Machinery Directive -1

Annex I, General Principles

… By the iterative process of risk assessment and risk reduction referred to above, the manufacturer or his authorized representative shall:

- …
- eliminate the hazards or reduce the risks associated with these hazards by application of protective measures, in the order of priority established in section 1.1.2(b).

2

Tab1-1/5-Sh.3

Three-step method

Machinery Directive, Annex I, chapter 1.1.2.b

Protective measures shall be applied in the following sequence, referred to as the three-step method:

1. Inherently safe machinery design and construction
2. Safeguarding and/or complementary protective measures
3. Information for use

Tab1-1/5-Sh.4

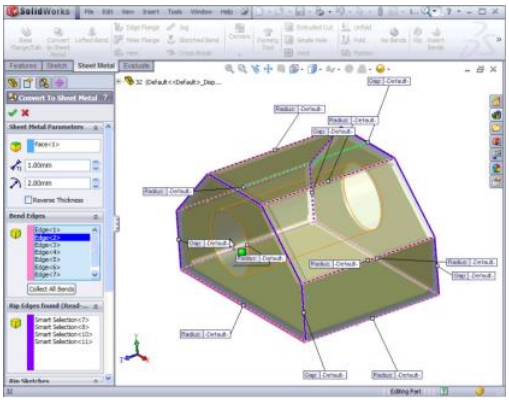
Question 3

- Which step does the emergency stop device belong to?

Inherently safe design measures

EN-ISO 12100, chapter 6.2

- Change design
- Eliminating hazards
- Reduce exposure



Inherently safe design measures

EN-ISO 12100

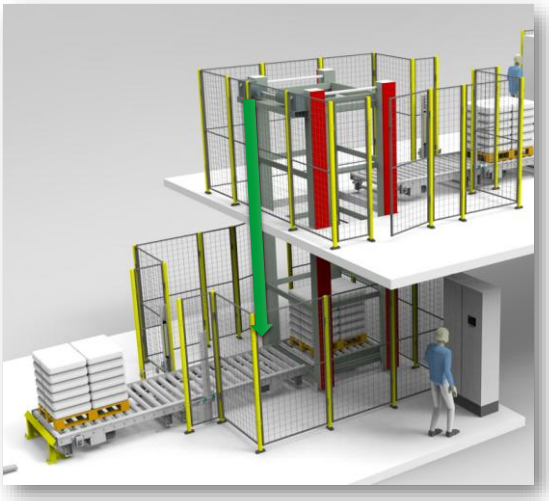
- Geometrical factors
  - avoiding sharp edges and corners
  - maximize direct visibility of the working areas and hazard zones from the control position
  - avoiding crushing increasing the minimum gap between the moving parts (or decreasing) EN-ISO 13854, EN-ISO 13857
- Physical aspects
  - decreasing force, velocity, mass, electric power
  - limiting the emissions by acting on the characteristics of the source:
    - reducing noise emissions ISO/TR 11688-1
    - reducing vibrations (hand-held machinery CR 1030-1)
    - reducing emission of hazardous substances
    - reducing radiation emissions
- General technical knowledge of machine design
  - stress limitation (methods)
  - selecting materials (corrosion, hardness, brittleness)
  - emission values for noise, vibrations, hazardous substances, radiation

Required sequence Step 1

EN-ISO 12100, chapter 6.2

Example of inherent measure (step 1).

- The pallet stacker shown is equipped with a drive at height (+/- 2m compared to floor area)
- The risk assessment shows that there is a significant risk related to the risk of falling for maintenance work on the drive.
- As an inherent measure, it can be considered to move the drive to the floor surface so that the specific risk of falling is completely eliminated.



Safeguarding and/or complementary protective measures

(EN-ISO 12100)

- Guards EN-ISO 14120
  - Fixed guards (transmissions, etc.)
  - Interlocking movable guard
  - Adjustable guards
- Fixed safe access devices EN ISO 14122-1/4
- Protective devices
  - Non-contact electrical protective devices EN 61496
  - Pressure-sensitive protective devices EN-ISO 13856
  - Two-hand control EN ISO 13851
- Protections and emissions (noise, vibration, radiation, dusts)

Safeguarding and/or complementary protective measures

(EN-ISO 12100)

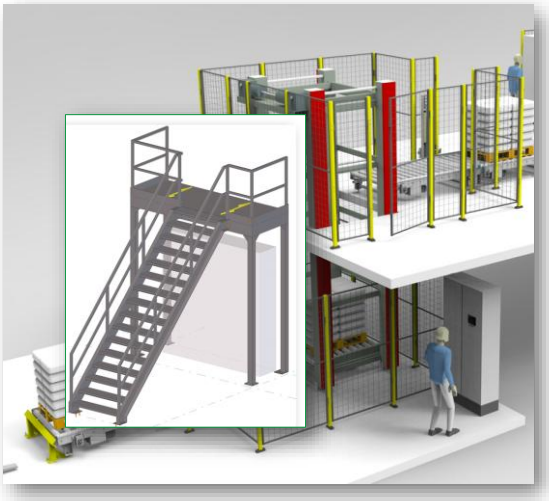
- Environmental protections (heat, cold)
- FOPS (Falling Object Protection)
- ROPS and TOPS (Protective Structure Against Tilting)
- Working within a protected area with protections disabled
- Safety distances (EN-ISO 13855, EN-ISO 13857)
- Safety measures to improve stability
- Motion limitation
- Acceleration/deceleration limitation
- Emergency stop devices (EN ISO 13850)

Required sequence step 2

EN-ISO 12100, chapter 6.3

Example Safeguarding and/or complementary protective measures (step 2)

- If the inherent measure from the previous example is not feasible, a landing with a fixed staircase can be provided to reach the relevant maintenance positions.
- Of course, the fencing and its integration must be taken into account.
- Therefore, investigate whether any new dangers may arise from this adjustment.....



Tab1-1/5-Sh.11

Three-step method step 3

EN-ISO 12100, chapter 6.4

- Information

Tab1-1/5-Sh.12

Information for use

EN-ISO 12100

- Markings, signs (pictograms) and written warnings
- function indication
- instruction handbook
- Training / instruction

Dwingende volgorde stap 3

EN-ISO 12100, hoofdstuk 6.4

Voorbeeld informatie voor gebruik (stap 3)

- Wanneer een machine voorzien is van hete delen waarbij het niet mogelijk is om:
  - de machine zo te ontwerpen dat met een veel lagere temperatuur gewerkt kan worden, en…
  - de hete delen niet afgeschermd kunnen worden,

- kan nabij de hete delen een waarschuwingspictogram voorzien worden. Neem daarbij dan ook in de gebruiksaanwijzing een waarschuwing op voor het gevaar en instructies hoe de gebruiker hier toch veilig kan werken.

Exercise 4-2

A company develops a pedestal drill machine as can be seen in the photo.

To which steps of the Three-step method do the following risk reducing measures belong:

- Interlocking guard (pully)
- Safety control system of interlocking guard
- Pictogram “safety glasses required”
- Emergency stop device
- Instructions accompanied by the drill press machine.

Sufficient risk reduction

EN-ISO 12100, chapter 6.2.4

Sufficient risk reduction is achieved if:

- All operating conditions have been taken into account
- Hazards have been eliminated or risks reduced to the lowest practical level
- Introduced hazards have also been assessed (and if necessary) reduced
- Residual risks are clearly described (users informed)
- Working conditions and serviceability of the machine intact
- Sufficient account has been taken of the use of an industrial machine in a NON-industrial environment (e.g. home)

Applying harmonised standards

Machinery Directive 1.3.7.
Risks related to the moving parts
The moving parts of machinery must be designed and constructed in such a way as to prevent the risk of contact which could lead to accidents or, where risks persist, be fitted with guards or protective devices.

Machinery Directive 1.4.1.
General requirements
Guards and protective devices:

- Must be sturdily executed,
- Must be held securely in place,
- Must not present additional hazards
- Must be sufficiently far from the danger zone,
- … etc

Applying harmonised standards

Height of hazard zone [c] *a*	Height of protective stru *b*					
	1 000	1 200	1 400	1 600	1 800	2 000
	Horizontal safety distance to ha					
2 700	0	0	0	0	0	0
2 600	900	800	700	600	600	500
2 400	1 100	1 000	900	800	700	600
2 200	1 300	1 200	1 000	900	800	600
2 000	1 400	1 300	1 100			
1 800	1 500	1 400	1 100			
1 600	1 500	1 400	1 100			
1 400	1 500	1 400	1 100			
1 200	1 500	1 400	1 100			

NEN-EN-ISO 13857
(en)

Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs (ISO 13857:2008,IDT)

Applying harmonised standards

Van Dale:

norm *(de; v(m);* meervoud: *normen)*

1 maatstaf, regel, richtsnoer: *naar een vaste norm*

richt·lijn *(de; v(m);* meervoud: *richtlijnen)*

1 aanwijzing voor te volgen gedrag

European Regulations:
Standard = Advice
Directive - directive = Law
Directive has 'legal force' once in National Legislation
Standard has 'legal force' (is evidence)

Applying harmonised standards

Why do you apply standards:

- Legal burden of proof (presumption of conformity)
- Also suitable as a state of the art with regard to the Work Equipment Directive (state of the art; framework directive article 6)
- Law gives vague requirement; Standards provide more concrete solutions

Standards mandatory?

Applying standards:

Where it says that the application of standards is not mandatory, but is recommended in order to meet the essential requirements of the Machinery Directive?

- Recital 18: '... Non-binding text..."
- Recital 11: '... a presumption of conformity..."

Tab1-1/5-Sh.21

Classification of standards

Type A standards:
- Fundamental safety principles
- *E.g. EN-ISO 12100 (e.g. risk assessment and risk reduction)*

Type B standards:
- Technical safety aspects and features
- *E.g. EN-IEC 60204-1 (electrical equipment of machines)*

Type C standards:
- Specific groups of machines
- *E.g. EN 619 (general cargo conveyors)*

Tab1-1/5-Sh.22

Question 4-3

What type of standards do the following belong?

- EN 13854: Minimum distances to prevent crushing?
- EN 614-1: Ergonomic design principles
- EN 1010-5: Corrugated board production machines and flat and corrugated board conversion machines
- EN-IEC 62061: Functional safety of safety-related control systems (SIL)
- EN-IEC 10218-1: Industrial robots

Question 4-4

- What tools are there to determine which standards apply to the risk assessment and reduction of a machine?

Presumption of conformity

- When is a standard harmonized?
- Which essential requirements does a specific standard give substance to??

EUROPEAN COMMISSION
Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs
Ecosystems III: Construction & machinery
Standards policy

Brussels, 4.5.2022

Summary of references of harmonised standards published in the Official Journal – Directive 2006/42/EC [1] of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC

The summary below consolidates the references of harmonised standards published by the Commission in the *Official Journal of the European Union* (OJ). It reproduces information already published in the L or C series of the OJ as indicated in columns (2), (5) and/or (7). It contains all references which, when the summary was generated, still provided a presumption of conformity together with references already withdrawn from the OJ.

The Commission services provide this summary for information purposes only. Although they take every possible precaution to ensure that the summary is updated regularly and is correct, errors may occur and the summary may not be complete at a certain point in time. The summary does not as such generate legal effects.

This summary was generated on 4 May 2022

Legislation reference (A)	ESO (B)	Reference number of the standard (C)	Title of the standard (D)	Type (E)	Date of start of presumption of conformity (1)	OJ reference for publication in OJ (2)	Restriction (3)	Date of start of presumption of conformity with restriction (4)	OJ reference for publication of a restriction in OJ (5)	Date of withdrawal from OJ (end of presumption of conformity) (6)	OJ reference for withdrawal from OJ (7)
2006/42/EC	CEN	EN ISO 12100:2010	Safety of machinery - General principles for design - Risk assessment and risk reduction (ISO 12100:2010)	A	08/04/2011	OJ C 110 - 08/04/2011	-		-		-

Periodical Journals European Commission

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of Directive 2006/42/EC machinery, and amending Directive 95/16/EC (recast) [2006 L157] aimed to be covered

This European Standard has been prepared under a Commission's standardization request M/396 to provide one voluntary means of conforming to essential requirements of Directive 2006/42/EC of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC (recast) [2006 L157].

Once this standard is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding essential requirements of that Directive, and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Directive 2006/42/EC [2006 L157]

Essential Requirements of Directive 2006/42/EC	Clause(s) / subclause(s) of this EN	Remarks / Notes
1.5.15 "Risks of slipping, tripping or falling"	All	
1.6.2 "Access to operating position and service points"	All	

Annex ZA, ZZA, ZB etc. in a European standard

Tab1-1/5-Sh.25

Presumption of conformity

When is a standard harmonized?

- In order to be able to harmonise a standard under a European directive, it must always be a European standard:
  - (NEN) **EN** ISO
  - (NEN) **EN** IEC
  - (NEN) **EN**
- There can be a difference between an IEC/ISO standard and the 'same' EN-IEC/ISO standard, example:
  - IEC 60204-1:2016
  - **EN** IEC 60204-1:2018
- In the documentation (TD) and statements, always use the year notation of the standard, for example "EN IEC 60204-1**:2018**

Tab1-1/5-Sh.26

Presumption of conformity

When is a standard harmonized?

- There is usually a transition period of about one year during which the harmonisation of an existing standard is transferred to that of the updated or replacement standard. Example:

Reference number of the standard (C)	Title of the standard (D)	Type (E)	Date of start of presumption of conformity (1)	OJ reference for publication in OJ (2)	Restriction (3)	Date of start of presumption of conformity with restriction (4)	OJ reference for publication of a restriction in OJ (5)	Date of withdrawal from OJ (end of presumption of conformity) (6)	OJ reference for withdrawal from OJ (7)
EN 62061:2005, EN 62061:2005/AC:2010, EN 62061:2005/A1:2013, EN 62061:2005/A2:2015	Safety of machinery - Functional safety of safety-related electrical, electronic and programmable electronic control systems	B	15/01/2016	OJ C 014 - 15/01/2016	-		-	11/10/2023	OJ L 115 - 13/04/2022
EN IEC 62061:2021	Safety of machinery - Functional safety of safety-related control systems	B	13/04/2022	OJ L 115 - 13/04/2022	-		-		-

Tab1-1/5-Sh.27

Exercise 4-9

- Briefly describe an appropriate risk-reducing measure for the identified risks to the pallet wrapper. Make use of harmonized standards as much as possible and refer to these standards in the control measure.
- Use the fill-in sheets at the back of Tab3 of the folder.
- Plenary discussion

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